Introduction to Option Pricing

Vadgama Devanshu Diljeet 20221088 Supervisor : Dr.Anindya Goswami

Spring 2026

Contents

1 Measure Theoretic Foundations **3**
1.1 Definitions . . . 3
1.2 Approximation by Simple Functions . . . 3
1.3 Convergence Theorems . . . 4

2 Probability Theory and Stochastic Processes **4**
2.1 Basic Definitions . . . 4
2.2 Limit Theorems and Laws . . . 4
2.3 Central Limit Theorem and Levy Continuity . . . 6

3 Construction and Existence Theorems **6**
3.1 Kolmogorov Consistency Theorem . . . 6
3.2 Kolmogorov-Chentsov Theorem . . . 8
3.3 Existence of Infinite Sequence of i.i.d. $U[0,1]$ . . . 8

4 Brownian Motion and Levy Processes **8**
4.1 Definitions . . . 8
4.2 Discussion on Normality and Continuity . . . 9

5 Fundamental Theorem of Asset Pricing (FTAP) **9**
5.1 Definitions . . . 9

6 Moments and Distributions **14**

7 Variation **15**

8 Itô Integral **19**
8.1 Defining Itô Integral for simple processes . . . 19
8.2 Extension of Itô Integral to General Processes . . . 20
8.3 Quadratic Variation and the Heuristic $dB_t^2 = dt$ . . . 21

9 European Option Pricing and Hedging **22**
9.1 Binomial Model: Replication and Pricing . . . 22
9.2 Hedging in the Binomial Model . . . 24
9.3 FTAP Part II . . . 24

10 Black–Scholes–Merton PDE **26**
10.1 Standard Approach . . . 26
10.2 Alternative Derivation via Discounted Processes . . . 27

11 Option Pricing **28**
11.1 Option Moneyness 28
11.2 Historical Volatility 28
11.3 Volatility in the Black–Scholes Model 29
11.4 Implied Volatility 29
11.5 Solution of the Geometric Brownian Motion 30
11.6 Derivation of the Black–Scholes Call Option Price 30
11.7 Breeden-Litzenberger Result 35
11.8 Dupire Local Volatility Formula 37

12 Derivation of the Brenner-Subrahmanyam Implied Volatility Formula **39**
12.1 Black-Scholes Model and ATMF Condition 39
12.2 Linear Approximation of the Cumulative Normal Distribution 39
12.3 Extension to Puts and Straddles 40

1 Measure Theoretic Foundations

1.1 Definitions

Definition 1.1 (σ-algebra)**.** *Let Ω be a non-empty set. A collection $\mathcal{F}$ of subsets of Ω is called a σ-algebra if:*

1. $\Omega \in \mathcal{F}$.
2. *If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$ (closed under complementation).*
3. *If $A_1, A_2, \dots \in \mathcal{F}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$ (closed under countable unions).*

The pair $(\Omega, \mathcal{F})$ is called a measurable space.

Definition 1.2 (Measurable Function)**.** *Let $(\Omega, \mathcal{F})$ and $(E, \mathcal{E})$ be measurable spaces. A function $f : \Omega \to E$ is measurable if for every $A \in \mathcal{E}$, the preimage $f^{-1}(A) = \{\omega \in \Omega : f(\omega) \in A\} \in \mathcal{F}$.*

Definition 1.3 (Integration in the Measure Theoretic Sense)**.** *Let $(\Omega, \mathcal{F}, \mu)$ be a measure space. The Lebesgue integral of a non-negative measurable function $f : \Omega \to [0, \infty]$ is defined as:*

$$\int_\Omega f \, d\mu = \sup \left\{ \int_\Omega \phi \, d\mu : 0 \le \phi \le f, \ \phi \text{ is simple} \right\}$$

where for a simple function $\phi = \sum_{i=1}^{n} a_i \mathbf{1}_{A_i}$, the integral is $\sum_{i=1}^{n} a_i \mu(A_i)$. For a general measurable function f, we define $\int f d\mu = \int f^+ d\mu - \int f^- d\mu$, provided at least one of the terms is finite.

1.2 Approximation by Simple Functions

Theorem 1.1 (Approximation Theorem)**.** *Let $f : \Omega \to [0, \infty]$ be a non-negative measurable function. Then there exists a sequence of simple measurable functions $\{\phi_n\}_{n \ge 1}$ such that $0 \le \phi_n \le \phi_{n+1}$ for all n, and $\lim_{n \to \infty} \phi_n(\omega) = f(\omega)$ for all $\omega \in \Omega$.*

Proof. For each $n \in \mathbb{N}$, define the function ϕ_n as follows:

$$\phi_n(\omega) = \sum_{k=0}^{n2^n - 1} \frac{k}{2^n} \mathbf{1}_{E_{n,k}}(\omega) + n \mathbf{1}_{F_n}(\omega)$$

where the sets are defined as:

$$E_{n,k} = \left\{ \omega \in \Omega : \frac{k}{2^n} \le f(\omega) < \frac{k+1}{2^n} \right\}, \quad F_n = \{\omega \in \Omega : f(\omega) \ge n\}$$

Since f is measurable, the sets $E_{n,k}$ and F_n are in $\mathcal{F}$, so ϕ_n is a simple measurable function.

1. **Monotonicity:** By construction, the partition at stage $n+1$ refines the partition at stage n (intervals of length 2^{-n} are split into two intervals of length $2^{-(n+1)}$). Thus, $\phi_n \le \phi_{n+1}$.
2. **Convergence:** If $f(\omega) < \infty$, for sufficiently large n (such that $n > f(\omega)$), $|\phi_n(\omega) - f(\omega)| < 2^{-n}$, which tends to 0. If $f(\omega) = \infty$, then $\phi_n(\omega) = n$ for all n, which tends to ∞.

Thus, $\phi_n \nearrow f$ pointwise. $\square$

1.3 Convergence Theorems

Theorem 1.2 (Monotone Convergence Theorem (MCT)). *Let $\{f_n\}$ be a sequence of non-negative measurable functions such that $f_n \uparrow f$ pointwise almost everywhere. Then:*

$$\lim_{n\to\infty} \int f_n \, d\mu = \int f \, d\mu$$

Theorem 1.3 (Dominated Convergence Theorem (DCT)). *Let $\{f_n\}$ be a sequence of measurable functions such that $f_n \to f$ pointwise almost everywhere. If there exists a measurable function $g \in L^1(\mu)$ (i.e., $\int |g| d\mu < \infty$) such that $|f_n| \leq g$ almost everywhere for all n, then $f \in L^1(\mu)$ and:*

$$\lim_{n\to\infty} \int f_n \, d\mu = \int f \, d\mu$$

2 Probability Theory and Stochastic Processes

2.1 Basic Definitions

Definition 2.1 (Filtration). *A filtration on a measurable space $(\Omega, \mathcal{F})$ is a sequence of σ-algebras $\{\mathcal{F}_t\}_{t\geq 0}$ such that $\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F}$ for all $s \leq t$. It represents the information available at time t.*

Definition 2.2 (Stochastic Process). *A stochastic process is a collection of random variables $\{X_t\}_{t\in T}$ defined on a common probability space $(\Omega, \mathcal{F}, \mathbb{P})$, indexed by a set T (usually representing time, e.g., $\mathbb{N}$ or $[0, \infty)$).*

Definition 2.3 (Martingales). *Let $\{X_t\}_{t\in T}$ be a stochastic process adapted to a filtration $\{\mathcal{F}_t\}$. Assuming $\mathbb{E}[|X_t|] < \infty$, the process is a:*

- ***Martingale*** *if $\mathbb{E}[X_t \mid \mathcal{F}_s] = X_s$ almost surely for all $s \leq t$.*
- ***Sub-martingale*** *if $\mathbb{E}[X_t \mid \mathcal{F}_s] \geq X_s$ almost surely for all $s \leq t$.*
- ***Super-martingale*** *if $\mathbb{E}[X_t \mid \mathcal{F}_s] \leq X_s$ almost surely for all $s \leq t$.*

2.2 Limit Theorems and Laws

Limit theorems play a fundamental role in probability theory and its applications to finance, insurance, and stochastic processes. They describe the long-run behavior of random sequences and allow us to make almost-sure or probabilistic conclusions about events that depend on infinitely many random variables.

Theorem 2.1 (Kolmogorov's 0–1 Law). *Let $\{X_n\}_{n\geq 1}$ be a sequence of independent random variables. Define the* tail σ-algebra

$$\mathcal{T} := \bigcap_{n=1}^{\infty} \sigma(X_n, X_{n+1}, \dots).$$

If an event $A \in \mathcal{T}$, then

$$\mathbb{P}(A) \in \{0, 1\}.$$

The tail σ-algebra consists of events that depend only on the asymptotic behavior of the sequence $\{X_n\}$ and are unaffected by any finite number of initial random variables. Kolmogorov's 0–1 law asserts that such events are *deterministic in probability*: they either occur almost surely or almost never.

This result is particularly useful in financial and insurance mathematics, where one is often interested in long-term outcomes. For example, events such as

- "Wealth grows exponentially in the long run",
- "An insurance company is eventually ruined",
- "A trading strategy succeeds infinitely often"

are tail events. Once it is shown that such an event has positive probability (or probability strictly less than one), Kolmogorov's 0–1 law immediately implies that the probability must in fact be equal to one (or zero, respectively).

Theorem 2.2 (Borel–Cantelli Lemmas)**.** *Let $\{A_n\}_{n\geq 1}$ be a sequence of events.*

- ***First Borel–Cantelli Lemma.*** *If*

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty,$$

then

$$\mathbb{P}\left(\limsup_{n\to\infty} A_n\right) = 0,$$

that is, the probability that infinitely many of the events A_n occur is zero.

- ***Second Borel–Cantelli Lemma.*** *If the events $\{A_n\}$ are independent and*

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty,$$

then

$$\mathbb{P}\left(\limsup_{n\to\infty} A_n\right) = 1.$$

Here,

$$\limsup_{n\to\infty} A_n := \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k$$

denotes the event that infinitely many of the events A_n occur.

The Borel–Cantelli lemmas provide sharp criteria for determining whether rare events occur only finitely many times or infinitely often. In financial contexts, these lemmas are used to analyze repeated market shocks, extreme losses, or default events over long time horizons. In particular, the second lemma highlights the crucial role of independence in ensuring that events with sufficiently large cumulative probability occur almost surely.

Together, Kolmogorov's 0–1 law and the Borel–Cantelli lemmas form essential tools for studying asymptotic phenomena in probability theory, underpinning results in risk theory, option pricing, and the long-term analysis of stochastic models.

2.3 Central Limit Theorem and Levy Continuity

Theorem 2.3 (Lindeberg-Feller CLT)**.** *Let $\{X_{n,k}\}$ be a triangular array of independent random variables with zero means and finite variances $\sigma_{n,k}^2$. Let $s_n^2 = \sum_{k=1}^{r_n} \sigma_{n,k}^2$. If the Lindeberg condition holds:*

$$\lim_{n\to\infty} \frac{1}{s_n^2} \sum_{k=1}^{r_n} \mathbb{E}\left[X_{n,k}^2 \cdot \mathbf{1}_{\{|X_{n,k}|>\epsilon s_n\}}\right] = 0 \quad \textit{for all } \epsilon > 0,$$

then $S_n/s_n \xrightarrow{d} \mathcal{N}(0,1)$.

Theorem 2.4 (Lévy Continuity Theorem)**.** *Let $\{F_n\}$ be a sequence of cumulative distribution functions, and for each n, define the characteristic function*

$$\varphi_n(t) := \mathbb{E}\big[e^{itX_n}\big] = \int_{\mathbb{R}} e^{itx}\, dF_n(x), \quad t \in \mathbb{R}.$$

Assume that

$$\varphi_n(t) \to \varphi(t) \quad \textit{for all } t \in \mathbb{R},$$

and that $\varphi(t)$ is continuous at $t = 0$. Then φ is the characteristic function of some random variable X, and

$$F_n \xrightarrow{d} F_X.$$

Theorem 2.5 (Strong Law of Large Numbers (SLLN))**.** *Let $X_1, X_2, \ldots$ be i.i.d. random variables with $\mathbb{E}[|X_1|] < \infty$ and $\mathbb{E}[X_1] = \mu$. Then $\frac{1}{n}\sum_{i=1}^n X_i \to \mu$ almost surely as $n \to \infty$.*

Theorem 2.6 (Carathéodory's Extension Theorem)**.** *Let X be a set and let $\mathcal{A}$ be an algebra of subsets of X. Suppose $\mu_0 : \mathcal{A} \to [0,\infty]$ is a pre-measure, that is,*

$$\mu_0(\varnothing) = 0$$

and

$$\mu_0\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu_0(A_n)$$

for every sequence $(A_n)_{n\geq 1} \subset \mathcal{A}$ of pairwise disjoint sets such that $\bigcup_{n=1}^{\infty} A_n \in \mathcal{A}$.

Then there exists a measure μ on the σ-algebra $\sigma(\mathcal{A})$ generated by $\mathcal{A}$ such that

$$\mu|_{\mathcal{A}} = \mu_0.$$

Moreover, if μ_0 is σ-finite on $\mathcal{A}$, then this extension is unique.

3 Construction and Existence Theorems

3.1 Kolmogorov Consistency Theorem

Theorem 3.1 (Kolmogorov Consistency Theorem)**.** *Let T be an index set. Suppose for every finite subset $t_1, \ldots, t_k \subset T$, we are given a probability measure $\mu_{t_1,\ldots,t_k}$ on $\mathbb{R}^k$. If the collection of measures satisfies two consistency conditions:*

1. ***Permutation Invariance.*** *Let μ be a probability measure on $\mathbb{R}^k$. For any permutation π of $\{1,\ldots,k\}$, define the mapping*

   $$T_\pi(x_1,\ldots,x_k) := (x_{\pi(1)},\ldots,x_{\pi(k)}).$$

   We say that μ is permutation invariant *if*

   $$\mu \circ T_\pi^{-1} = \mu \quad \textit{for all permutations } \pi.$$

   Equivalently, for every Borel set $A \subset \mathbb{R}^k$,

   $$\mu(A) = \mu\big(T_\pi^{-1}(A)\big).$$

2. ***Projection Consistency.*** *Let $\{\mu_{t_1,\dots,t_k}\}_{k\geq 1}$ be a family of probability measures on $\mathbb{R}^k$. We say that this family is* projection consistent *if for all $t_1,\dots,t_k,t_{k+1}$ and all Borel sets $A \in \mathcal{B}(\mathbb{R}^k)$,*

$$\mu_{t_1,\dots,t_k,t_{k+1}}(A \times \mathbb{R}) = \mu_{t_1,\dots,t_k}(A).$$

Then there exists a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and a stochastic process $X : T \times \Omega \to \mathbb{R}$ such that the finite dimensional distributions of X coincide with the given measures μ.

Proof Sketch. Let $\Omega = \mathbb{R}^T$ be the space of all functions from T to $\mathbb{R}$. We define cylinder sets of the form $\{\omega \in \Omega : (\omega(t_1),\dots,\omega(t_k)) \in B\}$ where $B \in \mathcal{B}(\mathbb{R}^k)$. Let $\mathcal{A}$ be the algebra generated by these cylinders. We define a set function $\mathbb{P}_0$ on $\mathcal{A}$ using the consistent measures $\mu_{t_1,\dots,t_k}$. The consistency conditions ensure $\mathbb{P}_0$ is well-defined. By Carathéodory's Extension Theorem, $\mathbb{P}_0$ extends uniquely to a measure $\mathbb{P}$ on $\sigma(\mathcal{A})$ if $\mathbb{P}_0$ is countably additive on $\mathcal{A}$. The proof of countable additivity typically relies on the topological regularity (compactness arguments) of measures on $\mathbb{R}^k$. □

Definition 3.1. *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let*

$$X = \{X_t : t \in T\}, \qquad Y = \{Y_t : t \in T\}$$

be two stochastic processes indexed by the same set T.

- ***Modification.*** *The process Y is said to be a modification of X if*

  $$\mathbb{P}(X_t = Y_t) = 1 \quad \text{for all } t \in T.$$

  That is, for each fixed time t, the random variables X_t and Y_t are equal almost surely. The exceptional null set may depend on t.

- ***Indistinguishability.*** *The processes X and Y are said to be indistinguishable if*

  $$\mathbb{P}\big(X_t = Y_t \text{ for all } t \in T\big) = 1.$$

  Equivalently, there exists a null set $N \subset \Omega$ such that $X_t(\omega) = Y_t(\omega)$ for all $t \in T$ and all $\omega \notin N$.

3.2 Kolmogorov-Chentsov Theorem

Theorem 3.2 (Kolmogorov-Chentsov Continuity Theorem)**.** *Let $X = \{X_t\}_{t\in[0,T]}$ be a stochastic process. If there exist positive constants α, β, C such that for all $s, t \in [0, T]$:*

$$\mathbb{E}[|X_t - X_s|^\alpha] \leq C|t - s|^{1+\beta}$$

Then there exists a modification $\tilde{X}$ of X which has continuous paths. Furthermore, the paths are locally Hölder continuous with exponent γ for any $0 < \gamma < \beta/\alpha$.

[1]

Proof Sketch. The proof utilizes the dyadic rationals $D = \bigcup_n D_n$ where $D_n = \{k2^{-n} : 0 \leq k \leq 2^n T\}$. By Chebyshev's inequality and the assumption:

$$\mathbb{P}(|X_{(k+1)2^{-n}} - X_{k2^{-n}}| > 2^{-n\gamma}) \leq \frac{E[|X_{(k+1)2^{-n}} - X_{k2^{-n}}|^\alpha]}{2^{-n\gamma\alpha}} \leq C\frac{(2^{-n})^{1+\beta}}{2^{-n\gamma\alpha}} = C2^{-n(1+\beta-\gamma\alpha)}$$

Summing over k and then over n, since $1 + \beta - \gamma\alpha > 1$, the series converges. By Borel-Cantelli Lemma 1, for almost all ω, $|X_t - X_s| \leq K(\omega)|t - s|^\gamma$ holds for all $s, t \in D$ sufficiently close. Since D is dense, we define $\tilde{X}_t(\omega) = \lim_{s\in D, s\to t} X_s(\omega)$. This $\tilde{X}$ is the continuous modification. □

3.3 Existence of Infinite Sequence of i.i.d. $U[0, 1]$

Question: Does there exist an infinite countable sequence of random variables $\{X_i\}$ such that $X_i \sim$ i.i.d. $U[0, 1]$? If so, construct one.

Rephrased Problem: Construct a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and a sequence of functions $X_n : \Omega \to [0, 1]$ such that the joint distribution of any finite subset is the product of uniform measures.

Answer: Yes. **Construction:** We can use the Kolmogorov Consistency Theorem on the space $\Omega = [0, 1]^{\mathbb{N}}$. However, a more explicit construction is possible using a single $U[0, 1]$ variable. Let $\omega \in (0, 1)$. Consider its binary expansion $\omega = 0.b_1b_2b_3 \cdots = \sum_{k=1}^{\infty} b_k 2^{-k}$. The digits b_k are i.i.d. Bernoulli(1/2). We can rearrange these bits into an infinite number of independent streams. Let p_n be the n-th prime. We can allocate bits to X_n based on indices divisible by powers of primes, or simply map the single sequence $b_1, b_2, \ldots$ into an infinite grid $b_{i,j}$ (using a bijection $\mathbb{N} \to \mathbb{N} \times \mathbb{N}$). Then define $X_n(\omega) = \sum_{j=1}^{\infty} b_{n,j}(\omega)2^{-j}$. These X_n will be i.i.d. $U[0, 1]$.

4 Brownian Motion and Levy Processes

4.1 Definitions

Definition 4.1 (Brownian Motion)**.** *A standard Brownian Motion is a stochastic process $B = \{B_t\}_{t\geq 0}$ satisfying:*

1. *$B_0 = 0$ almost surely.*
2. *B has independent, stationary increments. (For $0 \leq s < t$, $B_t - B_s$ is independent of $\mathcal{F}_s$ and its distribution depends only on $t - s$).*
3. *$B_t - B_s \sim \mathcal{N}(0, t - s)$.*
4. *The paths $t \mapsto B_t(\omega)$ are continuous almost surely.*

Definition 4.2 (Levy Process)**.** *A Levy process $X = \{X_t\}_{t\geq 0}$ is a stochastic process with:*

[1] f is Hölder continuous with exponent $\alpha \in (0, 1]$ if $\exists C > 0$ such that $|f(x) - f(y)| \leq C|x - y|^\alpha$ for all x, y.

1. *$X_0 = 0$ a.s.*
2. *Independent and stationary increments.*
3. *Stochastic continuity ($\lim_{t\to s} \mathbb{P}(|X_t - X_s| > \epsilon) = 0$).*

4.2 Discussion on Normality and Continuity

Question: If we drop the normality condition from Brownian Motion, does condition 2 (Stationary Indep. Increments) imply condition 3 (Normality) up to modification? What can you induce instead of Normal to preserve nature?

Analysis: If we drop Normality ($B_t - B_s \sim \mathcal{N}(0, t-s)$) but keep **Continuity** and **Stationary Independent Increments**, the process *must* be a Brownian Motion (possibly with drift and scaling). This is a result of the **Levy-Khintchine representation** for Levy processes. A Levy process can be decomposed into a drift, a Brownian component, and a jump component. If the paths are continuous, the jump component must be zero. Thus, **Continuity + Indep/Stat Increments $\implies$ Normality**.

However, if the question implies: "Does Stationary Independent Increments alone imply Continuity?", the answer is **No** (e.g., Poisson Process). Why is Normal special? By the Central Limit Theorem, the sum of many small independent shocks (random walk) converges to a Normal distribution. Brownian Motion is the scaling limit of Random Walks.

Brownian Motion as Limit of Random Walks: Let $S_n = \sum_{i=1}^n \xi_i$ where ξ_i are i.i.d. with mean 0 and variance 1. Define $W_t^{(n)} = \frac{1}{\sqrt{n}} S_{\lfloor nt \rfloor}$. By Donsker's Theorem (Functional CLT), $W^{(n)}$ converges in distribution to Brownian Motion B.

5 Fundamental Theorem of Asset Pricing (FTAP)

5.1 Definitions

Definition 5.1 (Market Setup)**.** *Consider a discrete time market $t = 0, \dots, T$ with $d+1$ assets.*

- *S_t^0: Risk-free asset (Bond). Assume $S_t^0 = (1+r)^t$.*
- *$S_t^1, \dots, S_t^d$: Risky assets.*

Definition 5.2 (Discounted Price Process)**.** *The discounted price process $\tilde{S}_t = S_t / S_t^0$.*

Definition 5.3 (Trading Strategy)**.** *A trading strategy is a predictable process $\phi = \{\phi_t\}_{t=1}^T$ where $\phi_t = (\alpha_t, \beta_t^1, \dots, \beta_t^d) \in \mathbb{R}^{d+1}$, ϕ_t represents the holdings during the interval $(t-1, t]$. Predictable means ϕ_t is $\mathcal{F}_{t-1}$-measurable.*

Definition 5.4 (Payoff/Value Process)**.** *The value of the portfolio at time t is $V_t(\phi) = \phi_t \cdot S_t$. The strategy is self-financing if $\phi_t \cdot S_t = \phi_{t+1} \cdot S_t$. The discounted gain process is $G_t(\phi) = \sum_{u=1}^t \beta_u \cdot (\tilde{S}_u - \tilde{S}_{u-1})$.*

Definition 5.5 (Arbitrage)**.** *An arbitrage opportunity is a self-financing strategy ϕ such that:*

$$V_0(\phi) = 0, \quad \mathbb{P}(V_T(\phi) \geq 0) = 1, \quad \textit{and} \quad \mathbb{P}(V_T(\phi) > 0) > 0$$

Definition 5.6 (Convex Hull)**.** *Let $C \subset \mathbb{R}^n$. The* convex hull *of C, denoted by* $\mathrm{conv}(C)$*, is defined as the set of all convex combinations of points in C, that is,*

$$\mathrm{conv}(C) := \left\{ \sum_{i=1}^m \lambda_i x_i \;\middle|\; x_i \in C,\ \lambda_i \geq 0,\ \sum_{i=1}^m \lambda_i = 1,\ m \in \mathbb{N} \right\}.$$

Equivalently, $\mathrm{conv}(C)$ *is the smallest convex set containing C.*

Theorem 5.1 (Strict Separating Hyperplane Theorem)**.** *Let C and K be two non-empty, disjoint, convex subsets of $\mathbb{R}^n$. Assume that C is compact and K is closed. Then there exists a non-zero vector $a \in \mathbb{R}^n$ and a real number $b \in \mathbb{R}$ such that*

$$\sup_{x \in C} \langle a, x \rangle < b < \inf_{y \in K} \langle a, y \rangle.$$

Equivalently, there exists a linear functional

$$L(x) := \langle a, x \rangle$$

that strictly separates *the sets C and K, meaning that*

$$L(x) < L(y) \quad \textit{for all } x \in C \textit{ and } y \in K.$$

Remark. Strict separation means that the two sets lie on opposite sides of the hyperplane

$$\{x \in \mathbb{R}^n : \langle a, x \rangle = b\},$$

with a positive gap between them. The assumptions that C is compact and K is closed ensure that the extrema of the linear functional are attained and that the separating hyperplane does not merely touch either set.

Theorem 5.2 (First Fundamental Theorem of Asset Pricing (FTAP))**.** *Consider a discrete–time market on a finite probability space $(\Omega, \mathcal{F}, P)$ with one risk–free asset S_0, d risky assets S_i^n, portfolio value process V_n, holdings ϕ_i^n in the ith risky asset at time n, and discounted prices denoted by $\tilde{S}_i^n$.*

Then :

$$\textit{The market admits no arbitrage} \quad \Longleftrightarrow \quad \textit{There exists an equivalent martingale measure } Q.$$

amssymb

Proof. **(EMM exists $\Rightarrow$ No Arbitrage).**

Assume an equivalent martingale measure Q exists. Let (ψ^n, ϕ^n) be any self-financing strategy with portfolio value process (V_n).

Step 1: Discounted portfolio is a Q-martingale.

From self-financing,

$$\tilde{V}_{n+1} - \tilde{V}_n = \sum_{i=1}^{d} \phi_i^n \big(\tilde{S}_i^{n+1} - \tilde{S}_i^n\big).$$

Taking conditional expectation under Q and using that ϕ_i^n is $\mathcal{F}_n$–measurable,

$$E_Q\Big[\tilde{V}_{n+1} - \tilde{V}_n \mid \mathcal{F}_n\Big] = \sum_{i=1}^{d} \phi_i^n E_Q\Big[\tilde{S}_i^{n+1} - \tilde{S}_i^n \mid \mathcal{F}_n\Big].$$

Since each $\tilde{S}_i^n$ is a Q–martingale,

$$E_Q[\tilde{S}_i^{n+1} \mid \mathcal{F}_n] = \tilde{S}_i^n.$$

Hence,

$$E_Q\Big[\tilde{S}_i^{n+1} - \tilde{S}_i^n \mid \mathcal{F}_n\Big] = 0,$$

and therefore

$$E_Q[\tilde{V}_{n+1} \mid \mathcal{F}_n] = \tilde{V}_n.$$

Thus $(\tilde{V}_n)$ is a Q-martingale.

Step 2: Relation between initial and terminal value.

From the martingale property,

$$\tilde{V}_0 = E_Q[\tilde{V}_N].$$

Step 3: Contradiction if arbitrage exists.

Suppose, for contradiction, that there exists an arbitrage strategy. Then

$$V_0 = 0, \qquad V_N \geq 0 \text{ a.s.}, \qquad P(V_N > 0) > 0.$$

Since $B_n > 0$,

$$\tilde{V}_N = \frac{V_N}{B_N} \geq 0 \quad \text{a.s.}$$

Because $Q \sim P$, equivalence of measures implies

$$Q(V_N > 0) > 0.$$

Hence $\tilde{V}_N$ is nonnegative and strictly positive with positive Q–probability. Therefore,

$$E_Q[\tilde{V}_N] > 0.$$

But from Step 2, $\tilde{V}_0 = E_Q[\tilde{V}_N]$. Since $V_0 = 0$ and $B_0 = 1$, we have $\tilde{V}_0 = 0$, so

$$0 = E_Q[\tilde{V}_N],$$

which is a contradiction.

Thus any arbitrage strategy cannot exist! Therefore, the existence of an equivalent martingale measure implies absence of arbitrage.

(No Arbitrage $\Rightarrow$ EMM).

To explain the main idea clearly, we present the construction in the one–period case; the general discrete–time case follows by applying the same argument at each step.

We work on a finite probability space $(\Omega, \mathcal{F}, P)$. There are d risky assets with prices S_i^0 at time 0 and $S_i^1(\omega)$ at time 1, and one risk–free asset with $B_0 = 1$, $B_1 = R > 0$. Define discounted prices

$$\tilde{S}_i^0 = S_i^0, \qquad \tilde{S}_i^1 = \frac{S_i^1}{R}.$$

A portfolio is a vector $\phi = (\phi_1, \ldots, \phi_d) \in \mathbb{R}^d$ with initial value

$$V_0 = \sum_{i=1}^{d} \phi_i S_i^0.$$

The discounted terminal value is

$$\tilde{V}_1(\omega) = \sum_{i=1}^{d} \phi_i \big(\tilde{S}_i^1(\omega) - S_i^0\big).$$

No–arbitrage condition.

If a portfolio ϕ satisfies

$$V_0 = 0 \quad \text{and} \quad \tilde{V}_1(\omega) \geq 0 \text{ for all } \omega \in \Omega,$$

then necessarily

$$\tilde{V}_1(\omega) = 0 \text{ for all } \omega.$$

Equivalently,

$$\phi \cdot (\tilde{S}_1(\omega) - S_0) \geq 0 \ \forall \omega \implies \phi \cdot (\tilde{S}_1(\omega) - S_0) = 0 \ \forall \omega.$$

Goal. Construct a probability measure Q such that

$$Q(\omega) > 0 \text{ for all } \omega \in \Omega, \qquad E_Q[\tilde{S}_i^1] = S_i^0 \quad \text{for } i = 1, \ldots, d.$$

Let $\Omega = \{\omega_1, \ldots, \omega_m\}$. We seek numbers $q_j > 0$ such that

$$\sum_{j=1}^{m} q_j = 1, \qquad \sum_{j=1}^{m} q_j \tilde{S}_i^1(\omega_j) = S_i^0, \quad i = 1, \ldots, d.$$

Equivalently,

$$\sum_{j=1}^{m} q_j \big(\tilde{S}_1(\omega_j) - S_0\big) = 0 \quad \text{in } \mathbb{R}^d.$$

Thus we want 0 to be a convex combination of the vectors

$$x_j := \tilde{S}_1(\omega_j) - S_0 \in \mathbb{R}^d.$$

Geometric argument.

We claim that 0 belongs to the convex hull of $\{x_1, \ldots, x_m\}$. Suppose not. Then by the separating hyperplane theorem, there exists $\phi \in \mathbb{R}^d$ such that

$$\phi \cdot x_j \geq 0 \text{ for all } j, \qquad \phi \cdot x_k > 0 \text{ for some } k.$$

This means

$$\phi \cdot (\tilde{S}_1(\omega) - S_0) \geq 0 \ \forall \omega,$$

and strictly positive for at least one state. Then the portfolio ϕ satisfies

$$V_0 = 0, \qquad \tilde{V}_1(\omega) = \phi \cdot (\tilde{S}_1(\omega) - S_0) \geq 0,$$

with strict positivity in at least one state, which is an arbitrage. This contradicts the no–arbitrage assumption.

Hence 0 must lie in the convex hull of $\{x_1, \ldots, x_m\}$. Therefore, there exist weights $q_1, \ldots, q_m$ such that

$$q_j > 0, \qquad \sum_{j=1}^{m} q_j = 1, \qquad \sum_{j=1}^{m} q_j x_j = 0.$$

That is,

$$\sum_{j=1}^{m} q_j \big(\tilde{S}_1(\omega_j) - S_0\big) = 0,$$

which implies

$$\sum_{j=1}^{m} q_j \tilde{S}_1(\omega_j) = S_0.$$

Define a probability measure Q by

$$Q(\{\omega_j\}) = q_j.$$

Then

$$E_Q[\tilde{S}_1] = \sum_{j=1}^{m} q_j \tilde{S}_1(\omega_j) = S_0,$$

and since all $q_j > 0$, we have $Q \sim P$. Hence the discounted asset prices are martingales under Q, so Q is an equivalent martingale measure. □

NOTE : Why the one–period argument implies the N–period case?

In a multi–period discrete–time market, trading over N periods can be viewed as a sequence of one–period transitions. Absence of arbitrage over the whole horizon implies absence of arbitrage between any two consecutive times n and $n+1$. Applying the one–period result at each step gives a consistent system of strictly positive probabilities that makes the one–step discounted price increments have zero conditional mean. Piecing these one–step constructions together yields a single probability measure Q under which each discounted price process $(\tilde{S}_i^n)$ is a martingale for all $n = 0, \ldots, N$. Thus the one–period construction extends to the general discrete–time model.

NOTE: EMM Need Not Be Unique

The equivalent martingale measure (EMM) need not be unique. Non–uniqueness occurs when the number of possible future states exceeds the number of risky assets, leading to an incomplete market.

Example. Consider a one–period model with one risky asset and one risk–free asset:

$$B_0 = 1, \quad B_1 = R > 0, \quad S_0 = 1.$$

Suppose the risky asset has three possible values at time 1:

$$S_1(\omega_1) = 0.8, \quad S_1(\omega_2) = 1.2, \quad S_1(\omega_3) = 1.6.$$

Let $Q(\omega_j) = q_j > 0$ with $q_1 + q_2 + q_3 = 1$. The martingale condition

$$E_Q[\tilde{S}_1] = \tilde{S}_0 = 1$$

gives

$$0.8q_1 + 1.2q_2 + 1.6q_3 = R.$$

This is one equation together with $q_1 + q_2 + q_3 = 1$. Since there are three unknowns and only two equations, infinitely many positive solutions (q_1, q_2, q_3) exist. Each solution defines a different EMM.

Thus the EMM is not unique. This happens because one asset cannot uniquely determine probabilities across three states. In such cases, the market is incomplete and some contingent claims do not have a unique arbitrage–free price.
[2]

[2] A risk–neutral measure (also called an equivalent martingale measure) is a probability measure Q equivalent to P such that all discounted asset price processes $\tilde{S}_i^n$ are martingales under Q, i.e.

$$E_Q[\tilde{S}_i^{n+1} \mid \mathcal{F}_n] = \tilde{S}_i^n \quad \text{for all } i, n.$$

Equivalently, under Q, the expected return of each asset equals the risk–free rate when measured in discounted terms.

6 Moments and Distributions

Definition 6.1 (Kurtosis)**.** *Kurtosis is a measure of the* tailedness *of the probability distribution of a real-valued random variable. For a random variable X with mean μ and standard deviation σ, the kurtosis is defined as*

$$\mathbb{E}\left[\left(\frac{X-\mu}{\sigma}\right)^4\right].$$

The excess kurtosis *is defined as the kurtosis minus* 3*, which corresponds to the kurtosis of a normal distribution.*

In financial applications, particularly in equity and derivative markets, random variables often represent *log-returns* of asset prices. Empirical evidence shows that financial return distributions frequently deviate from normality, exhibiting skewness and heavy tails. These deviations play a crucial role in option pricing and risk management.

Interpretation of the first four moments:

- **First Moment (Mean):** Measures the central tendency. In finance, this corresponds to the average return of an asset over a given time horizon.
- **Second Moment (Variance):** Measures dispersion or spread. This is directly related to volatility, a key input in option pricing models such as Black–Scholes.
- **Third Moment (Skewness):** Measures asymmetry of the distribution. Negative skewness is commonly observed in equity returns, reflecting the higher probability of large downward moves.
- **Fourth Moment (Kurtosis):** Measures the heaviness of the tails and the presence of outliers. High kurtosis indicates a greater likelihood of extreme price movements, which leads to underestimation of risk when assuming normally distributed returns.

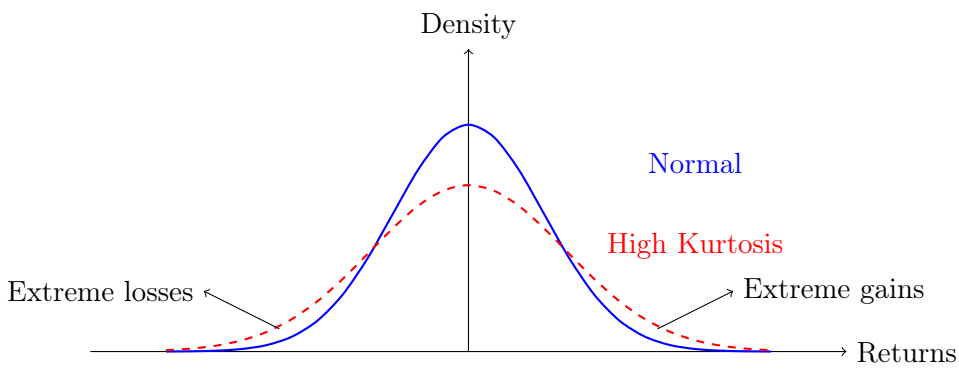

Figure 1: Illustration of return distributions in financial markets. Compared to the normal distribution, empirical stock return distributions exhibit heavier tails, reflecting higher kurtosis and increased probability of extreme price movements.

In the context of option pricing, higher kurtosis implies that extreme market events occur more frequently than predicted by Gaussian models. This explains phenomena such as volatility smiles and motivates the use of alternative models incorporating jumps or stochastic volatility.

7 Variation

Total Variation. Let $f : [a, b] \to \mathbb{R}$. The total variation of f on $[a, b]$ is

$$V_a^b(f) = \sup_P \sum_{i=1}^{n} \big|f(t_i) - f(t_{i-1})\big|,$$

where the supremum is taken over all partitions $P = \{a = t_0 < t_1 < \cdots < t_n = b\}$. If $V_a^b(f) < \infty$, then f is said to be of bounded variation.

Quadratic Variation. Let $X = (X_t)_{t\in[0,T]}$ be a real-valued process. Its quadratic variation is defined by

$$[X]_t = \lim_{|P|\to 0} \sum_{i=1}^{n} \big(X_{t_i} - X_{t_{i-1}}\big)^2 = \lim_{n\to\infty} \sum_{i=1}^{n} \big(X_{t_i} - X_{t_{i-1}}\big)^2,$$

whenever the limit exists and $|P| \to 0$ as $n \to \infty$, where $P = \{0 = t_0 < t_1 < \cdots < t_n = t\}$ denotes a partition of the interval $[0, t]$ and $|P| = \max_i(t_i - t_{i-1})$ represents the mesh size of the partition. For stochastic processes, this limit is understood in probability[3].

p-th variation (measurable sense). Let $X = (X_t)_{t\in[0,T]}$ be a real-valued measurable process. For a partition $\Pi = \{0 = t_0 < t_1 < \cdots < t_n = t\}$, define

$$V^{(p)}(X, \Pi)_t = \sum_{i=0}^{n-1} \big|X_{t_{i+1}} - X_{t_i}\big|^p, \qquad p > 0.$$

If there exists a measurable process $[X]_t^{(p)}$ such that for every sequence of partitions (Π_n) with mesh $|\Pi_n| \to 0$,

$$V^{(p)}(X, \Pi_n)_t \xrightarrow[n\to\infty]{} [X]_t^{(p)} \quad \text{in probability,}$$

then $[X]_t^{(p)}$ is called the *p-th variation* of X on $[0, t]$.

Proposition 7.1. *Let $f : [0, t] \to \mathbb{R}$ be a function of bounded variation on $[0, t]$. Then its quadratic variation on $[0, t]$ is zero, i.e.*

$$[f]_t = \lim_{|\Pi|\to 0} \sum_i \big(f(t_{i+1}) - f(t_i)\big)^2 = 0.$$

Proof. Let $\Pi = \{0 = t_0 < t_1 < \cdots < t_n = t\}$ be any partition of $[0, t]$. Then

$$\sum_{i=0}^{n-1} (f(t_{i+1}) - f(t_i))^2 \le \Big(\max_i |f(t_{i+1}) - f(t_i)|\Big) \sum_{i=0}^{n-1} |f(t_{i+1}) - f(t_i)|.$$

Since f has bounded variation,

$$\sum_{i=0}^{n-1} |f(t_{i+1}) - f(t_i)| \le V_t(f) < \infty.$$

[3] A sequence of random variables Y_n converges in probability to Y if for every $\varepsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}(|Y_n - Y| > \varepsilon) = 0.$$

Moreover, as $|\Pi| \to 0$, the increments satisfy

$$\max_i |f(t_{i+1}) - f(t_i)| \to 0.$$

Hence,

$$\sum_{i=0}^{n-1} (f(t_{i+1}) - f(t_i))^2 \leq V_t(f) \max_i |f(t_{i+1}) - f(t_i)| \longrightarrow 0 \quad \text{as } |\Pi| \to 0.$$

Therefore,

$$[f]_t = \lim_{|\Pi| \to 0} \sum_{i=0}^{n-1} (f(t_{i+1}) - f(t_i))^2 = 0.$$

□

Proposition 7.2. *Let $(B_t)_{t \in [0,T]}$ be a Brownian motion. Then, almost surely,*

1. *the quadratic variation exists and satisfies*

$$[B]_t = t, \qquad t \in [0, T],$$

2. *the paths of B_t have unbounded total variation on every interval $[0, t]$, $t > 0$.*

Proof. Fix $t \in [0, T]$ and consider the uniform partition

$$\Pi_n = \left\{0, \tfrac{t}{n}, \tfrac{2t}{n}, \ldots, t\right\}.$$

Set

$$Q_n(t) = \sum_{i=0}^{n-1} \left(B_{t_{i+1}} - B_{t_i}\right)^2, \qquad t_i = \frac{it}{n}.$$

Since Brownian motion has independent increments and

$$B_{t_{i+1}} - B_{t_i} \sim \mathcal{N}\left(0, \frac{t}{n}\right),$$

we have

$$\mathbb{E}\left[(B_{t_{i+1}} - B_{t_i})^2\right] = \frac{t}{n}.$$

Hence

$$\mathbb{E}[Q_n(t)] = \sum_{i=0}^{n-1} \frac{t}{n} = t.$$

Moreover, using independence and the fact that for $Z \sim \mathcal{N}(0, \sigma^2)$, $\mathrm{Var}(Z^2) = 2\sigma^4$, we obtain

$$\mathrm{Var}(Q_n(t)) = \sum_{i=0}^{n-1} 2\left(\frac{t}{n}\right)^2 = \frac{2t^2}{n} \longrightarrow 0.$$

Thus $Q_n(t) \to t$ in L^2, hence in probability, and along a subsequence almost surely. Since the limit is deterministic, this implies

$$[B]_t = \lim_{n \to \infty} Q_n(t) = t \quad \text{a.s.}$$

To prove that the total variation is infinite, consider

$$V_n(t) = \sum_{i=0}^{n-1} \left| B_{t_{i+1}} - B_{t_i} \right|.$$

Each increment is distributed as $|\mathcal{N}(0, t/n)|$, and

$$\mathbb{E}\big[|B_{t_{i+1}} - B_{t_i}|\big] = \sqrt{\frac{t}{n}}\, \mathbb{E}|Z| = \sqrt{\frac{t}{n}}\, \sqrt{\frac{2}{\pi}}, \quad Z \sim \mathcal{N}(0,1).$$

Therefore,

$$\mathbb{E}[V_n(t)] = n\sqrt{\frac{t}{n}}\sqrt{\frac{2}{\pi}} = \sqrt{\frac{2t}{\pi}}\, \sqrt{n} \longrightarrow \infty.$$

Hence $V_n(t) \to \infty$ in probability, and along a subsequence almost surely. Since the total variation dominates these sums over partitions,

$$\sup_{\Pi} \sum_i |B_{t_{i+1}} - B_{t_i}| \;\geq\; \limsup_{n\to\infty} V_n(t) = \infty \quad \text{a.s.}$$

Thus Brownian paths have unbounded total variation on $[0, t]$ for every $t > 0$. □

Implications of $[B]_t = t$ a.s. and infinite total variation. Let $(B_t)_{t\in[0,T]}$ be Brownian motion. The facts

$$[B]_t = t \quad \text{a.s.} \qquad \text{and} \qquad V_t(B) = \infty \quad \text{a.s.}$$

have the following fundamental consequences:

1. **Brownian paths are not of bounded variation.** Since bounded variation implies zero quadratic variation, and $[B]_t = t > 0$, Brownian motion cannot have finite total variation.

2. **Brownian paths are nowhere differentiable (a.s.).** If a function is differentiable on $[0, t]$, then it has bounded variation on compact intervals. Since Brownian paths have infinite variation, they cannot be differentiable.

3. **Classical calculus does not apply.** The Riemann–Stieltjes integral $\int f\, dB$ is not well-defined in general because the integrator (B) must have bounded variation. This necessitates the construction of the Itô integral.

4. **Quadratic variation drives stochastic calculus.** The identity $[B]_t = t$ produces the correction term in Itô's formula:

$$d(B_t^2) = 2B_t\, dB_t + dt.$$

5. **Roughness characterization.** Brownian motion is a canonical example of a process with:

   infinite total variation but finite quadratic variation.

   This distinguishes it sharply from smooth or finite-variation paths.

6. **Martingale structure.** The process $B_t^2 - t$ is a martingale, directly reflecting the fact that $[B]_t = t$.

7. **Scaling intuition.** Over an interval of size Δt,

$$|B_{t+\Delta t} - B_t| \sim \sqrt{\Delta t},$$

so

$$\sum |\Delta B| \sim \sqrt{n} \to \infty, \qquad \sum (\Delta B)^2 \to t.$$

Thus the path is too rough for total variation but regular enough for quadratic variation to stabilize.

Question 7.1. Is the map $t \mapsto [B]_t$ an identity map?

Answer. No. More precisely, it is true that

$$[B]_t = t \quad \text{a.s. for all } t \in [0, T],$$

but this does not mean that for every $\omega \in \Omega$ the path $t \mapsto [B]_t(\omega)$ is the identity map.

Explanation. By definition of quadratic variation, there exists a null set $N \subset \Omega$ with $\mathbb{P}(N) = 0$ such that for every $\omega \in \Omega \setminus N$ and every $t \in [0, T]$,

$$[B]_t(\omega) = t.$$

Hence, for each fixed t, the random variable $[B]_t$ satisfies

$$\mathbb{P}\big([B]_t = t\big) = 1.$$

However, this does *not* imply that

$$[B]_t(\omega) = t \quad \text{for all } t \text{ and all } \omega \in \Omega.$$

Indeed, there may exist exceptional outcomes $\omega \in N$ (a null set) for which the limit defining the quadratic variation either fails to hold or does not equal t for some t.

Thus, the statement should be interpreted pathwise as: there exists a set $\Omega_0 \subset \Omega$ with $\mathbb{P}(\Omega_0) = 1$ such that for every $\omega \in \Omega_0$,

$$t \longmapsto [B]_t(\omega)$$

is exactly the identity function on $[0, T]$.

But on the complement $N = \Omega \setminus \Omega_0$ (a null set), the equality may fail. Therefore, as a map on the whole probability space, $t \mapsto [B]_t$ is not literally the identity; it coincides with the identity only almost surely.

8 Itô Integral

8.1 Defining Itô Integral for simple processes

Let $(B_t)_{t\geq 0}$ be a standard Brownian motion on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$. Consider a simple process

$$S_t = \sum_{i=0}^{n-1} \xi_i \mathbf{1}_{[t_i, t_{i+1})}(t),$$

where $0 = t_0 < t_1 < \cdots < t_n = T$ is a partition of $[0, T]$ and each ξ_i is a random variable (r.v.) on Ω that is $\mathcal{F}_{t_i}$-measurable.

Why ξ_i Must Be $\mathcal{F}_{t_i}$-Measurable

Requiring ξ_i to be $\mathcal{F}_{t_i}$-measurable ensures that it depends only on the information available up to time t_i, i.e., the process is *non-anticipating*.

- **Random variable property:** Each ξ_i is a r.v. on Ω, meaning it is a measurable mapping $\xi_i : \Omega \to \mathbb{R}$. This allows us to perform expectations and variance computations.
- **No "peeking into the future":** If ξ_i depended on $B_{t_{i+1}}$ or later values, the integral would use future information, violating the definition of the Itô integral.
- **Independence from future increments:** Since $B_{t_{i+1}} - B_{t_i}$ is independent of $\mathcal{F}_{t_i}$, we can factor expectations when computing moments:

$$\mathbb{E}[\xi_i(B_{t_{i+1}} - B_{t_i})] = \mathbb{E}[\xi_i] \cdot \mathbb{E}[B_{t_{i+1}} - B_{t_i}] = 0,$$

$$\mathbb{E}[\xi_i^2(B_{t_{i+1}} - B_{t_i})^2] = \mathbb{E}[\xi_i^2]\,(t_{i+1} - t_i).$$

Itô Integral and Second Moment

The Itô integral of S with respect to B is defined as

$$\int_0^T S_t \, dB_t := \sum_{i=0}^{n-1} \xi_i (B_{t_{i+1}} - B_{t_i}).$$

Using the independence and Gaussian properties of Brownian increments, we compute the second moment:

$$\begin{aligned}
\mathbb{E}\left[\left(\int_0^T S_t \, dB_t\right)^2\right] &= \mathbb{E}\left[\left(\sum_{i=0}^{n-1} \xi_i (B_{t_{i+1}} - B_{t_i})\right)^2\right] \\
&= \sum_{i=0}^{n-1} \mathbb{E}[\xi_i^2]\,\mathbb{E}[(B_{t_{i+1}} - B_{t_i})^2] \\
&= \sum_{i=0}^{n-1} \mathbb{E}[\xi_i^2]\,(t_{i+1} - t_i) \\
&= \mathbb{E}\left[\int_0^T S_t^2 \, dt\right].
\end{aligned}$$

This gives the fundamental Itô isometry for simple processes:

$$\mathbb{E}\Big[\Big(\int_0^T S_t \, dB_t\Big)^2\Big] = \mathbb{E}\Big[\int_0^T S_t^2 \, dt\Big].$$

8.2 Extension of Itô Integral to General Processes

Let $(B_t)_{t\geq 0}$ be a standard Brownian motion on $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$. We now discuss how the Itô integral

$$\int_0^T f_t \, dB_t$$

can be extended from simple processes to a general adapted process f_t. The extension is done in three steps. We will asume

$$\mathbb{E}\Big[\int_0^T f_t^2 \, dt\Big] < \infty$$

1. Continuous Adapted Processes

Suppose f_t is continuous and $\mathcal{F}_t$-adapted. We can approximate f_t uniformly by a sequence of simple processes $(S_t^n)_{n\geq 1}$:

$$S_t^n := \sum_{i=0}^{k_n-1} \xi_i^n \mathbf{1}_{[t_i^n, t_{i+1}^n)}(t),$$

where ξ_i^n is $\mathcal{F}_{t_i^n}$-measurable. Then we define

$$\int_0^T f_t \, dB_t := \lim_{n\to\infty} \int_0^T S_t^n \, dB_t,$$

where the limit is taken in $L^2(\Omega)$ (mean square sense). The Itô isometry ensures convergence:

$$\mathbb{E}\Big[\Big(\int_0^T (f_t - S_t^n) \, dB_t\Big)^2\Big] = \mathbb{E}\Big[\int_0^T (f_t - S_t^n)^2 \, dt\Big] \to 0.$$

2. Bounded Adapted Processes

If f_t is bounded and $\mathcal{F}_t$-adapted (but not necessarily continuous), we can approximate it in $L^2([0,T]\times\Omega)$ by simple processes (S_t^n):

$$\int_0^T f_t \, dB_t := \lim_{n\to\infty} \int_0^T S_t^n \, dB_t,$$

again with convergence in $L^2(\Omega)$. Boundedness ensures the dominated convergence theorem applies, so the limit is well-defined.

3. Unbounded Adapted Processes

Finally, for a general $\mathcal{F}_t$-adapted process f_t satisfying

$$\mathbb{E}\Big[\int_0^T f_t^2 \, dt\Big] < \infty,$$

we first truncate f_t to define a bounded sequence

$$f_t^n := f_t \mathbf{1}_{\{|f_t|\leq n\}}.$$

Each f_t^n is bounded and adapted, so its Itô integral is well-defined. We then define

$$\int_0^T f_t \, dB_t := \lim_{n\to\infty} \int_0^T f_t^n \, dB_t,$$

with convergence in $L^2(\Omega)$. The Itô isometry guarantees that the limit exists and is independent of the particular truncation or approximation sequence.

Summary

Thus, starting from simple processes, the Itô integral can be consistently extended to any $\mathcal{F}_t$-adapted process f_t that is square-integrable:

$$f \in L^2([0,T] \times \Omega) \quad \Longrightarrow \quad \int_0^T f_t \, dB_t \text{ is well-defined in } L^2(\Omega).$$

Note that we won't be dealing with r.vs whose variance is infinite.

8.3 Quadratic Variation and the Heuristic $dB_t^2 = dt$

Let B_t be a standard Brownian motion and f_t an adapted process satisfying

$$\mathbb{E}\Big[\int_0^T f_t^2 \, dt\Big] < \infty.$$

Consider a partition $0 = t_0 < t_1 < \cdots < t_n = T$ and the simple approximation

$$S_t = \sum_{i=0}^{n-1} f_{t_i} \mathbf{1}_{[t_i, t_{i+1})}(t).$$

The Itô integral of S_t satisfies the Itô isometry:

$$\mathbb{E}\Big[\Big(\int_0^T S_t \, dB_t\Big)^2\Big] = \mathbb{E}\Big[\int_0^T S_t^2 \, dt\Big].$$

Expanding the left-hand side:

$$\mathbb{E}\Big[\Big(\sum_{i=0}^{n-1} f_{t_i}(B_{t_{i+1}} - B_{t_i})\Big)^2\Big] = \sum_{i=0}^{n-1} \mathbb{E}[f_{t_i}^2]\,(t_{i+1} - t_i) + \sum_{i \neq j} \mathbb{E}[f_{t_i} f_{t_j}(B_{t_{i+1}} - B_{t_i})(B_{t_{j+1}} - B_{t_j})].$$

By independence of increments, the cross terms vanish:

$$\mathbb{E}[(B_{t_{i+1}} - B_{t_i})(B_{t_{j+1}} - B_{t_j})] = 0 \quad \text{for } i \neq j.$$

Hence

$$\mathbb{E}\Big[\Big(\sum_{i=0}^{n-1} f_{t_i}(B_{t_{i+1}} - B_{t_i})\Big)^2\Big] = \sum_{i=0}^{n-1} \mathbb{E}[f_{t_i}^2]\,(t_{i+1} - t_i) \approx \mathbb{E}\Big[\int_0^T f_t^2 \, dt\Big].$$

Taking the limit as the mesh of the partition goes to zero, this shows that the "quadratic variation" of B_t satisfies

$$[B]_T := \lim_{\|\pi\| \to 0} \sum_{i=0}^{n-1} (B_{t_{i+1}} - B_{t_i})^2 = T \quad \text{almost surely,}$$

which rigorously justifies the heuristic differential identity

$$(dB_t)^2 = dt$$

in stochastic calculus.

9 European Option Pricing and Hedging

9.1 Binomial Model: Replication and Pricing

Let the payoff be

$$H = g(S_N),$$

where H is $\mathcal{F}_N$-measurable.

In the one-period binomial model, at time $n-1$ the stock price is S_{n-1}. At time n, it moves to

$$S_n = \begin{cases} uS_{n-1} \\ dS_{n-1} \end{cases}$$

Assume a self-financing portfolio with holdings ϕ_n^1 in stock and ϕ_n^0 in the bank account. Then,

If the stock goes up:

$$V_n(uS_{n-1}) = \phi_n^1 uS_{n-1} + \phi_n^0(1+r),$$

If the stock goes down:

$$V_n(dS_{n-1}) = \phi_n^1 dS_{n-1} + \phi_n^0(1+r).$$

Solving the two equations,

$$\phi_n^1 = \frac{V_n(uS_{n-1}) - V_n(dS_{n-1})}{(u-d)S_{n-1}},$$

$$\phi_n^0 = \frac{uV_n(dS_{n-1}) - dV_n(uS_{n-1})}{(u-d)(1+r)}.$$

The portfolio value at time $n-1$ is

$$V_{n-1} = \phi_n^1 S_{n-1} + \phi_n^0. \quad \text{(by self financing strategy)}$$

Given a terminal payoff $V_N = H$, backward induction recursively determines (ϕ_n^0, ϕ_n^1) and the entire price process $(V_n)_{n=0}^N$. Thus every contingent claim is replicated by a self-financing strategy, and the binomial market is complete.

—

Risk-Neutral Measure

Stock dynamics:

$$S_n = s\prod_{i=1}^{n} \xi_i,$$

where

$$\xi_i = \begin{cases} u & \text{with probability } p \\ d & \text{with probability } 1-p. \end{cases}$$

No-Arbitrage Condition

$$u \geq 1 + r \geq d.$$

Define risk-neutral probability q such that

$$\frac{suq + sd(1-q)}{s} = 1 + r.$$

i.e,

$$q = \frac{1 + r - d}{u - d}.$$

Under Q,

$$Q\left(\prod_{i=1}^{n} \xi_i = u^j d^{n-j}\right) = \binom{n}{j} q^j (1-q)^{n-j}.$$

Fair price of $H = g(S_N)$ (Fair Price is Expectation under optimal strategy):

$$H_0 = (1+r)^{-N} \sum_{j=0}^{N} g(su^j d^{N-j}) \binom{N}{j} q^j (1-q)^{N-j}.$$

For European call:

$$g(x) = \max(x - K, 0).$$

For European put:

$$g(x) = \max(K - x, 0).$$

—

Payoff: A payoff is the (possibly random) amount of money received at a specified future time, typically maturity T. Mathematically, it is a measurable function $H = g(S_T)$ defined on the underlying probability space.

Contingent Claim: A contingent claim is a financial contract whose payoff depends on the outcome of an underlying asset or random event at maturity. Formally, it is an $\mathcal{F}_T$-measurable random variable H.

Complete Market: A financial market is said to be complete if every contingent claim can be replicated exactly by a self-financing trading strategy. Equivalently, a market is complete if there exists a unique equivalent martingale measure.

Option: An option is a financial derivative that gives its holder the right, but not the obligation, to buy or sell an underlying asset at a predetermined price (strike price) on or before a specified maturity date.

Call Option: A call option gives its holder the right, but not the obligation, to purchase the underlying asset at the strike price K at maturity T. Its payoff is

$$(S_T - K)^+ = \max(S_T - K, 0).$$

Put Option: A put option gives its holder the right, but not the obligation, to sell the underlying asset at the strike price K at maturity T. Its payoff is

$$(K - S_T)^+ = \max(K - S_T, 0).$$

Option Buyer (Holder): The option buyer, also called the holder or long position, pays a premium at initiation and acquires the contractual right specified by the option.

Option Writer (Seller): The option writer, also called the seller or short position, receives the premium at initiation and assumes the obligation to fulfill the contract if the holder exercises the option.

9.2 Hedging in the Binomial Model

Suppose we want to obtain a payoff

$$H = g(S_N)$$

at maturity N. We begin by setting $V_N = H$ and construct a self-financing trading strategy in the stock and the risk-free asset that replicates this payoff.

At time $n-1$, the stock price is S_{n-1} and at time n it can move either to uS_{n-1} or dS_{n-1}. We choose portfolio holdings ϕ_n^1 (stock) and ϕ_n^0 (risk-free asset) so that the portfolio matches the required value in both possible future states:

$$\phi_n^1 uS_{n-1} + \phi_n^0(1+r) = V_n(uS_{n-1}),$$

$$\phi_n^1 dS_{n-1} + \phi_n^0(1+r) = V_n(dS_{n-1}).$$

These two equations ensure that, regardless of whether the stock moves up or down, the portfolio value at time n coincides with the desired continuation value of the claim.

The value of the portfolio at time $n-1$ is then

$$V_{n-1} = \phi_n^1 S_{n-1} + \phi_n^0,$$

which represents the amount of capital required at time $n-1$ to guarantee the future payoff.

By repeating this argument backward from time N to time 0, we determine both the entire hedging strategy and the initial capital V_0 needed to replicate H. Thus the claim is perfectly hedged, and since every contingent claim can be replicated in this manner, the binomial market is complete and the no-arbitrage price equals the replication cost.

Suppose we want to obtain a payoff $H = g(S_N)$ at maturity N. In the binomial model, we construct a self-financing trading strategy in the stock and the risk-free asset so that the portfolio value at time N exactly equals the desired payoff, regardless of how the stock price evolves.

At each time step, we choose the portfolio holdings so that the value of the portfolio in the next period matches the required value in both the up and down states. By repeating this procedure backward from maturity to time 0, we determine both the entire trading strategy and the initial capital required.

This procedure is called hedging because the constructed portfolio eliminates risk: no matter how the stock moves, the final payoff is guaranteed. Since every contingent claim can be replicated in this way, the binomial model is complete, and the no-arbitrage price of the claim is exactly the initial cost of the replicating strategy.

9.3 FTAP Part II

Theorem 9.1. *In a finite discrete-time arbitrage-free market, the market is complete if and only if the equivalent martingale measure is unique.*

Proof. ($\Rightarrow$) Suppose the market is complete. Then every contingent claim H is attainable, so there exists a self-financing strategy with initial capital V_0 such that $V_N = H$.

Under any equivalent martingale measure Q, discounted portfolio values are martingales. Hence for every such Q,

$$V_0 = \mathbb{E}^Q\left[\frac{H}{(1+r)^N}\right].$$

If there were two different equivalent martingale measures Q_1 and Q_2, then for some contingent claim H we would have

$$\mathbb{E}^{Q_1}\left[\frac{H}{(1+r)^N}\right] \neq \mathbb{E}^{Q_2}\left[\frac{H}{(1+r)^N}\right],$$

which would imply two different prices for H, contradicting completeness. Therefore, the equivalent martingale measure must be unique.

($\Leftarrow$) Conversely, suppose the equivalent martingale measure Q is unique. In a finite state space, prices are given by the linear functional

$$H \mapsto \mathbb{E}^Q\left[\frac{H}{(1+r)^N}\right].$$

Uniqueness of Q implies that this pricing functional is uniquely determined.

In finite-dimensional spaces, uniqueness of the pricing functional implies that the span of attainable payoffs equals the entire space of contingent claims. Hence every contingent claim is attainable, and the market is complete.

□

Prerequisites for the Black–Scholes–Merton Model.

The Black–Scholes–Merton (BSM) framework is formulated in continuous time and relies on tools from stochastic calculus. Unlike the binomial model, where trading occurs at discrete dates $n = 0, 1, \ldots, N$, a continuous-time market allows trading at every instant $t \in [0, T]$. Asset prices are therefore modeled as stochastic processes indexed by continuous time.

The randomness in the model is driven by a Brownian motion (or Wiener process) $(W_t)_{t\geq 0}$, which is a continuous-time stochastic process with independent, normally distributed increments and $W_0 = 0$. The stock price process (S_t) is assumed to satisfy a stochastic differential equation (SDE) of the form

$$dS_t = \mu S_t\, dt + \sigma S_t\, dW_t,$$

where μ is the drift (expected rate of return) and $\sigma > 0$ is the volatility , where volatility is standard deviation of returns,i.e, if R represents the return , $R = \frac{dS_t}{S_t}$ then $\sigma = \sqrt{(Var(R)}$. The term dW_t represents the random shock over an infinitesimal time interval, while dt denotes deterministic time evolution.

A trading strategy in continuous time consists of processes (η_t, ψ_t), where η_t denotes the number of shares held in the stock at time t, and ψ_t denotes the amount invested in the risk-free asset. The risk-free asset (money market account) evolves according to

$$dB_t = rB_t\, dt,$$

where r is the constant risk-free interest rate. The portfolio value is therefore

$$V_t = \eta_t S_t + \psi_t B_t.$$

A strategy is said to be self-financing if changes in portfolio value arise only from changes in asset prices and not from additional capital injection.

The fundamental analytical tool in the BSM derivation is Itô's Lemma. If S_t satisfies an SDE and $f(t, S_t)$ is a sufficiently smooth function, then Itô's Lemma provides the differential of $f(t, S_t)$:

$$df(t, S_t) = \frac{\partial f}{\partial t}dt + \frac{\partial f}{\partial x}dS_t + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}\sigma^2 S_t^2 dt.$$

The additional second-order term arises from the quadratic variation of Brownian motion, namely $(dW_t)^2 = dt$. We can casually use

	dt	dW_t
dt	0	0
dW_t	0	dt

10 Black–Scholes–Merton PDE

10.1 Standard Approach

Consider a trading strategy consisting of holding η_t units of the stock and ψ_t units in the risk-free asset. The portfolio value is

$$V_t = \eta_t S_t + \psi_t B_t.$$

Assume the stock price follows the stochastic differential equation

$$dS_t = \mu S_t dt + \sigma S_t dW_t,$$

and the risk-free asset evolves as

$$dB_t = rB_t dt.$$

If the strategy is self-financing, then changes in portfolio value arise only from changes in asset prices, so

$$dV_t = \eta_t dS_t + \psi_t dB_t.$$

Substituting the dynamics of S_t and B_t, we obtain

$$dV_t = \eta_t(\mu S_t dt + \sigma S_t dW_t) + \psi_t(rB_t dt).$$

Now let the value of a contingent claim be given by a sufficiently smooth function $V(t, S_t)$. Applying Itô's Lemma,

$$dV = \frac{\partial V}{\partial t}dt + \frac{\partial V}{\partial S}dS_t + \frac{1}{2}\frac{\partial^2 V}{\partial S^2}(dS_t)^2.$$

Since $(dS_t)^2 = \sigma^2 S_t^2 dt$, we obtain

$$dV = \left(\frac{\partial V}{\partial t} + \mu S_t \frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2}\right) dt + \sigma S_t \frac{\partial V}{\partial S} dW_t.$$

We now construct a portfolio that eliminates the stochastic term by choosing

$$\eta_t = \frac{\partial V}{\partial S}.$$

Then the dW_t term vanishes, and the portfolio becomes locally riskless. Since it is riskless, it must earn the risk-free rate:

$$dV = rV dt.$$

Equating the drift terms gives

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} + rS_t \frac{\partial V}{\partial S} - rV = 0.$$

This is the Black–Scholes–Merton partial differential equation.

10.2 Alternative Derivation via Discounted Processes

(Courtesy of Supervisor :))

Assume the stock price satisfies

$$dS_t = \mu S_t \, dt + \sigma S_t \, dW_t,$$

and the risk-free asset evolves as

$$dB_t = rB_t \, dt, \qquad B_t = e^{rt}.$$

Define the discounted stock price

$$\tilde{S}_t = \frac{S_t}{B_t}.$$

$$d\tilde{S}_t = \frac{1}{B_t} dS_t - \frac{S_t}{B_t^2} dB_t = \frac{S_t}{B_t}(\mu - r) \, dt + \frac{\sigma S_t}{B_t} \, dW_t.$$

Now let $V(t, S_t)$ denote the price of a contingent claim, and define its discounted value

$$\tilde{V}_t = \frac{V(t, S_t)}{B_t}.$$

Applying Itô's Lemma to $V(t, S_t)$ gives

$$dV = \left(\frac{\partial V}{\partial t} + \mu S_t \frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} \right) dt + \sigma S_t \frac{\partial V}{\partial S} \, dW_t.$$

Since,

$$d\tilde{V}_t = \frac{1}{B_t} dV - \frac{r}{B_t} V \, dt.$$

Substituting the expression for dV, we obtain

$$d\tilde{V}_t = \frac{1}{B_t} \left(\frac{\partial V}{\partial t} + \mu S_t \frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} - rV \right) dt + \frac{\sigma S_t}{B_t} \frac{\partial V}{\partial S} \, dW_t.$$

In an arbitrage-free market, discounted asset prices are martingales under the risk-neutral measure. Therefore, the drift term of $\tilde{V}_t$ must vanish. Hence,

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} + rS_t \frac{\partial V}{\partial S} - rV = 0.$$

11 Option Pricing

11.1 Option Moneyness

- **In the Money (ITM)**
  - **Call Option:** A call option is in the money when the underlying asset price is greater than the strike price.
$$S > K$$
  - **Put Option:** A put option is in the money when the underlying asset price is less than the strike price.
$$S < K$$
- **At the Money (ATM)**
  - **Call Option:** $S = K$
  - **Put Option:** $S = K$
- **Out of the Money (OTM)**
  - **Call Option:** A call option is out of the money when the underlying asset price is less than the strike price.
$$S < K$$
  - **Put Option:** A put option is out of the money when the underlying asset price is greater than the strike price.
$$S > K$$

11.2 Historical Volatility

Historical volatility measures the variability of the underlying asset price based on past market data. It is typically estimated using the standard deviation of the asset's log returns.

Let S_t denote the asset price at time t. The continuously compounded return is defined as

$$r_t = \ln\left(\frac{S_t}{S_{t-1}}\right)$$

Given n observations of returns $\{r_1, r_2, \ldots, r_n\}$, the sample mean return is

$$\bar{r} = \frac{1}{n}\sum_{i=1}^{n} r_i$$

The historical volatility (standard deviation of returns) is then estimated as

$$\sigma_{\text{hist}} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(r_i - \bar{r})^2}$$

If the data frequency is daily, the annualized historical volatility is

$$\sigma_{\text{annual}} = \sigma_{\text{hist}}\sqrt{252}$$

where 252 is the approximate number of trading days in a year.

11.3 Volatility in the Black–Scholes Model

In the Black–Scholes model, the underlying asset price is assumed to follow a geometric Brownian motion given by

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

where

- μ is the drift of the asset,
- σ is the volatility of the asset,
- W_t is a standard Brownian motion.

Applying Itô's lemma to $\ln S_t$, we obtain

$$d(\ln S_t) = \left(\mu - \frac{1}{2}\sigma^2\right) dt + \sigma dW_t$$

Integrating from 0 to T gives

$$\ln\left(\frac{S_T}{S_0}\right) = \left(\mu - \frac{1}{2}\sigma^2\right) T + \sigma W_T$$

Since $W_T \sim N(0, T)$, it follows that

$$\ln\left(\frac{S_T}{S_0}\right) \sim N\left(\left(\mu - \frac{1}{2}\sigma^2\right) T, \sigma^2 T\right)$$

Thus, in the Black–Scholes framework the asset price follows a lognormal distribution and the parameter σ represents the volatility of the asset's continuously compounded returns.

11.4 Implied Volatility

In the Black–Scholes model, the price of an option depends on several observable quantities such as the current asset price S, strike price K, time to maturity T, and the risk–free interest rate r. However, the volatility σ of the underlying asset is not directly observable in the market.

Instead, volatility is inferred from observed option prices. The value of σ that makes the Black–Scholes price equal to the market price of the option is called the *implied volatility*.

Let C_{market} denote the observed price of a European call option. The implied volatility σ_{imp} is defined implicitly by

$$C_{\text{market}} = C_{BS}(S, K, T, r, \sigma_{\text{imp}})$$

where C_{BS} denotes the Black–Scholes pricing formula. Since σ enters the formula nonlinearly, it cannot be solved for analytically and must be obtained using numerical methods such as the Newton–Raphson or bisection method.

Implied volatility differs from historical volatility in both interpretation and construction. Historical volatility is computed from past asset price movements and therefore reflects realized variability of returns. In contrast, implied volatility is extracted from option prices and represents the market's expectation of future volatility. Consequently, historical volatility is a backward-looking measure, whereas implied volatility is forward-looking.

Remark. Although the Black–Scholes model assumes a constant volatility parameter σ, empirical option prices imply different volatilities for different strike prices. When plotted as a function of the strike, these implied volatilities typically exhibit patterns known as the *volatility smile* or *volatility skew*.

11.5 Solution of the Geometric Brownian Motion

Assume the stock price follows the stochastic differential equation

$$dS_t = rS_t\,dt + \sigma S_t\,dW_t$$

where r is the risk-free rate, σ is the volatility, and W_t is a standard Brownian motion.
To solve this SDE, apply Itô's lemma to

$$f(S_t) = \ln S_t$$

Using Itô's lemma,

$$df = \left(\frac{\partial f}{\partial S} rS_t + \frac{1}{2}\frac{\partial^2 f}{\partial S^2}\sigma^2 S_t^2\right) dt + \frac{\partial f}{\partial S}\sigma S_t dW_t$$

Compute the derivatives:

$$\frac{\partial f}{\partial S} = \frac{1}{S_t}, \qquad \frac{\partial^2 f}{\partial S^2} = -\frac{1}{S_t^2}$$

Substituting,

$$d(\ln S_t) = \left(r - \frac{1}{2}\sigma^2\right) dt + \sigma dW_t$$

$$\int_t^T d(\ln S_s) = \int_t^T \left(r - \frac{1}{2}\sigma^2\right) ds + \int_t^T \sigma dW_s$$

$$\ln S_T - \ln S_t = \left(r - \frac{1}{2}\sigma^2\right)(T-t) + \sigma(W_T - W_t)$$

Since

$$W_T - W_t \sim N(0, T-t)$$

we obtain

$$S_T = S_t \exp\left[\left(r - \frac{1}{2}\sigma^2\right)(T-t) + \sigma N(0, T-t)\right]$$

Rewriting this ,

$$S_T = S_t \exp\left[\left(r - \frac{1}{2}\sigma^2\right)(\tau) + \sigma\sqrt{\tau} Z\right]$$

where,

$$Z \sim \mathcal{N}(0,1)\ ,\ \tau = T - t$$

11.6 Derivation of the Black–Scholes Call Option Price

Under the risk–neutral measure Q, the price of a European call option is

$$C(S_t, t) = e^{-r\tau}\mathbb{E}^Q[(S_T - K)^+ \mid S_t]$$

where $\tau = T - t$.
Since

$$(S_T - K)^+ = (S_T - K)\mathbf{1}_{\{S_T > K\}},$$

we write

$$C(S_t, t) = e^{-r\tau}\mathbb{E}^Q[(S_T - K)\mathbf{1}_{\{S_T > K\}} \mid S_t].$$

This becomes

$$C(S_t, t) = e^{-r\tau}\left(\mathbb{E}^Q[S_T\mathbf{1}_{\{S_T > K\}}|S_t] - K\,\mathbb{E}^Q[\mathbf{1}_{\{S_T > K\}}|S_t]\right).$$

Distribution of S_T
From the GBM solution,

$$\ln S_T = \ln S_t + (r - \tfrac{1}{2}\sigma^2)\tau + \sigma\sqrt{\tau}\, Z,$$

where $Z \sim N(0,1)$.
Thus

$$S_T = S_t \exp\left((r - \tfrac{1}{2}\sigma^2)\tau + \sigma\sqrt{\tau}Z\right).$$

Event transformation
Consider the event $\{S_T > K\}$.

$$S_t \exp\left((r - \tfrac{1}{2}\sigma^2)\tau + \sigma\sqrt{\tau}Z\right) > K$$

Taking logarithms,

$$\ln S_t + (r - \tfrac{1}{2}\sigma^2)\tau + \sigma\sqrt{\tau}Z > \ln K.$$

Hence

$$Z > \frac{\ln(K/S_t) - (r - \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}$$

Define

$$d_2 = \frac{\ln(S_t/K) + (r - \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}.$$

Then

$$\{S_T > K\} = \{Z > -d_2\}.$$

Second expectation

$$\mathbb{E}^Q[\mathbf{1}_{\{S_T>K\}}|S_t] = P(Z > -d_2)$$

$$= \int_{-d_2}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz$$

$$= N(d_2).$$

First expectation

$$\mathbb{E}^Q[S_T \mathbf{1}_{\{S_T>K\}}|S_t]$$

$$= \int_{-d_2}^{\infty} S_t \exp\left((r - \tfrac{1}{2}\sigma^2)\tau + \sigma\sqrt{\tau}z\right) \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz.$$

Factor constants:

$$= S_t e^{(r-\frac{1}{2}\sigma^2)\tau} \int_{-d_2}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left(\sigma\sqrt{\tau}z - \frac{z^2}{2}\right) dz.$$

Complete the square:

$$\sigma\sqrt{\tau}z - \frac{z^2}{2} = -\frac{1}{2}(z^2 - 2\sigma\sqrt{\tau}z) = -\frac{1}{2}(z - \sigma\sqrt{\tau})^2 + \frac{1}{2}\sigma^2\tau.$$

Thus

$$= S_t e^{(r-\frac{1}{2}\sigma^2)\tau} e^{\frac{1}{2}\sigma^2\tau} \int_{-d_2}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-(z-\sigma\sqrt{\tau})^2/2} dz.$$

$$= S_t e^{r\tau} \int_{-d_2}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-(z-\sigma\sqrt{\tau})^2/2} dz.$$

Let

$$y = z - \sigma\sqrt{\tau}.$$

Then the lower limit becomes

$$-d_2 - \sigma\sqrt{\tau} = -d_1$$

where

$$d_1 = \frac{\ln(S_t/K) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}.$$

Hence

$$\mathbb{E}^Q[S_T \mathbf{1}_{\{S_T > K\}} | S_t] = S_t e^{r\tau} \int_{-d_1}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-y^2/2} dy$$

$$= S_t e^{r\tau} N(d_1).$$

Substitute back

$$C(S_t, t) = e^{-r\tau} \left(S_t e^{r\tau} N(d_1) - K N(d_2) \right).$$

Therefore

$$\boxed{C(S_t, t) = S_t N(d_1) - K e^{-r\tau} N(d_2)}$$

Derivation of the Black–Scholes Formula via the Heat Equation

The Black–Scholes PDE is

$$\frac{\partial C}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} + rS\frac{\partial C}{\partial S} - rC = 0$$

with terminal condition

$$C(S, T) = \max(S - K, 0).$$

Time Transformation

Let

$$\tau = T - t$$

Then

$$\frac{\partial C}{\partial t} = -\frac{\partial C}{\partial \tau}$$

Substituting,

$$\frac{\partial C}{\partial \tau} = \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} + rS\frac{\partial C}{\partial S} - rC$$

with initial condition

$$C(S, 0) = \max(S - K, 0).$$

Log Price Transformation

Define

$$x = \ln\left(\frac{S}{K}\right)$$

and write

$$C(S,t) = K\,v(x,\tau).$$

Using the chain rule,

$$\frac{\partial C}{\partial S} = \frac{K}{S}\frac{\partial v}{\partial x}$$

$$\frac{\partial^2 C}{\partial S^2} = \frac{K}{S^2}\left(\frac{\partial^2 v}{\partial x^2} - \frac{\partial v}{\partial x}\right)$$

Substituting into the PDE gives

$$\frac{\partial v}{\partial \tau} = \frac{1}{2}\sigma^2\left(\frac{\partial^2 v}{\partial x^2} - \frac{\partial v}{\partial x}\right) + r\frac{\partial v}{\partial x} - rv$$

which simplifies to

$$\frac{\partial v}{\partial \tau} = \frac{1}{2}\sigma^2\frac{\partial^2 v}{\partial x^2} + \left(r - \frac{1}{2}\sigma^2\right)\frac{\partial v}{\partial x} - rv$$

Removing the First Derivative

Let

$$v(x,\tau) = e^{ax+b\tau}u(x,\tau)$$

Substituting and choosing constants a, b so that the first derivative disappears gives

$$a = \frac{1}{2} - \frac{r}{\sigma^2}, \qquad b = -\frac{1}{2}\sigma^2 a^2 - r$$

which transforms the PDE into

$$\frac{\partial u}{\partial \tau} = \frac{1}{2}\sigma^2\frac{\partial^2 u}{\partial x^2}.$$

This is the standard heat equation.

Initial Condition

From the payoff

$$C(S,T) = \max(S-K,0)$$

we obtain

$$v(x,0) = \max(e^x - 1, 0).$$

Using the transformation,

$$u(x,0) = e^{-ax}\max(e^x - 1, 0).$$

Solution of the Heat Equation

The heat equation solution is

$$u(x,\tau) = \frac{1}{\sqrt{2\pi\sigma^2\tau}} \int_{-\infty}^{\infty} u(y,0) \exp\left(-\frac{(x-y)^2}{2\sigma^2\tau}\right) dy.$$

Evaluating this convolution integral and transforming back to $C(S,t)$ yields

$$\boxed{C_{BS}(S,t) = S\,N(d_1) - Ke^{-r(T-t)}N(d_2)}$$

where

$$d_1 = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}}$$

$$d_2 = d_1 - \sigma\sqrt{T-t}.$$

Calculating Implied Volatility

After obtaining the theoretical price of a European call option using the Black–Scholes formula, we can determine the implied volatility from observed market prices.

Let C_{mkt} denote the observed market price of the option. The **implied volatility** is defined as the value σ_{imp} such that

$$C_{BS}(S,K,r,T,\sigma_{imp}) = C_{mkt}.$$

Equivalently, we define the function

$$f(\sigma) = C_{BS}(S,K,r,T,\sigma) - C_{mkt}.$$

The implied volatility is the root of the equation

$$f(\sigma) = 0.$$

Since the Black–Scholes formula cannot be inverted analytically for σ, numerical methods are used to compute the implied volatility.

A common method is the **Newton–Raphson iteration**. Starting from an initial guess σ_0, the iteration is given by

$$\sigma_{n+1} = \sigma_n - \frac{f(\sigma_n)}{f'(\sigma_n)}.$$

Here

$$f(\sigma) = C_{BS}(\sigma) - C_{mkt}$$

and

$$f'(\sigma) = \frac{\partial C}{\partial \sigma}.$$

The derivative of the option price with respect to volatility is known as the **Vega**. For a European call option,

$$\text{Vega} = \frac{\partial C}{\partial \sigma} = S\sqrt{T-t}\,\phi(d_1)$$

(notice that $\implies Vega >= 0$)

where

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}.$$

Thus the Newton–Raphson iteration becomes

$$\sigma_{n+1} = \sigma_n - \frac{C_{BS}(\sigma_n) - C_{mkt}}{S\sqrt{T-t}\,\phi(d_1)}.$$

This procedure is repeated until convergence, yielding the implied volatility σ_{imp}. The implied volatility represents the volatility parameter that makes the Black–Scholes price equal to the observed market price of the option. Newton Raphson generally fails on real data when either Vega is close to zero (deep ITM or deep OTM) and also when we close to maturity ie $T \approx t$. So we Brent's Method to get teh implied volatility.

11.7 Breeden-Litzenberger Result

Consider the risk–neutral pricing formula for a European call option with strike K and maturity T:

$$C(K) = e^{-r\tau}\mathbb{E}^Q[(S_T - K)^+]$$

where $\tau = T - t$. Let $f_{S_T}(s)$ denote the risk–neutral probability density of S_T. The expectation can be written as

$$C(K) = e^{-r\tau}\int_0^\infty (s-K)^+ f_{S_T}(s)\,ds.$$

Since

$$(s-K)^+ = \begin{cases} s-K & s > K, \\ 0 & s \le K, \end{cases}$$

the integral becomes

$$C(K) = e^{-r\tau}\int_K^\infty (s-K) f_{S_T}(s)\,ds.$$

Differentiating with respect to K,

$$\frac{\partial C}{\partial K} = e^{-r\tau}\frac{\partial}{\partial K}\int_K^\infty (s-K) f_{S_T}(s)\,ds.$$

Using Leibnitz's rule,

$$\frac{\partial C}{\partial K} = e^{-r\tau}\left[-\int_K^\infty f_{S_T}(s)\,ds\right].$$

Thus,

$$\frac{\partial C}{\partial K} = -e^{-r\tau}Q(S_T > K).$$

Differentiating once more,

$$\frac{\partial^2 C}{\partial K^2} = -e^{-r\tau}\frac{\partial}{\partial K}Q(S_T > K).$$

Since

$$\frac{\partial}{\partial K}Q(S_T > K) = -f_{S_T}(K),$$

we obtain

$$\boxed{\frac{\partial^2 C}{\partial K^2} = e^{-r\tau} f_{S_T}(K)}$$

Hence the risk–neutral density can be recovered from option prices as

$$f_{S_T}(K) = e^{r\tau}\frac{\partial^2 C}{\partial K^2}.$$

Estimating the Risk–Neutral Density from Market Option Data

The Breeden–Litzenberger result shows that the risk–neutral density of the terminal asset price can be recovered from call option prices via

$$f_{S_T}(K) = e^{r\tau} \frac{\partial^2 C(K)}{\partial K^2}.$$

In practice, market data provides call prices only at a discrete set of strikes, so the derivatives must be approximated numerically. The procedure is as follows.

Step 1: Collect Option Market Data

Obtain market data for European call options with the same maturity T. For each strike K_i, collect:

$$(K_i, C_i)$$

where C_i is the observed market price of the call option.

If option prices are not directly used, implied volatilities may first be converted to call prices using the Black–Scholes formula.

Step 2: Construct a Smooth Call Price Function

Since option prices are observed only at discrete strikes, construct a smooth function

$$C(K)$$

that interpolates the observed prices. Common methods include:

- cubic spline interpolation
- Richardson Extrapolation

This ensures that the second derivative with respect to K can be computed reliably.

Step 3: Compute the Second Derivative

Once the smooth function $C(K)$ is obtained, compute

$$\frac{\partial^2 C}{\partial K^2}$$

either analytically from the fitted function or numerically using finite differences. For discrete strikes K_{i-1}, K_i, K_{i+1}, the central difference approximation is

$$\frac{\partial^2 C}{\partial K^2}(K_i) \approx \frac{C_{i+1} - 2C_i + C_{i-1}}{(\Delta K)^2}.$$

Step 4: Recover the Risk–Neutral Density

Using the Breeden–Litzenberger formula,

$$f_{S_T}(K_i) = e^{r\tau} \frac{\partial^2 C}{\partial K^2}(K_i).$$

Repeating this for all strikes provides an estimate of the entire risk–neutral density.

Step 5: Normalize the Density

Numerical errors may cause the density to deviate from a proper probability distribution. Therefore normalize so that

$$\int_0^\infty f_{S_T}(s)\, ds = 1.$$

Volatility Smile :): Discovery and Motivation

In the Black–Scholes model, volatility σ is assumed to be constant and independent of the strike price and maturity of the option. Under this assumption, if market option prices are converted into implied volatilities using the Black–Scholes formula, all options with the same maturity should produce the same implied volatility. Thus, implied volatility should be constant across strikes.

However, empirical observations in financial markets show that this is not the case. When traders compute implied volatilities from observed option prices across different strike prices, the resulting values vary systematically with the strike. When these implied volatilities are plotted as a function of the strike price, the curve often resembles a "smile" shape. This phenomenon is known as the **volatility smile**.

Historical Discovery

The volatility smile was first widely observed after the stock market crash of 1987. Prior to this event, market participants often treated the Black–Scholes model as an accurate description of option prices. After the crash, traders noticed that deep out-of-the-money put options traded at significantly higher prices than predicted by the Black–Scholes model.

When these prices were converted to implied volatilities, they produced much higher volatility estimates for low strikes compared to at-the-money options. This pattern indicated that the assumption of constant volatility was inconsistent with market data.

Interpretation

The volatility smile reflects the fact that the risk-neutral distribution of asset prices implied by the market is not lognormal, as assumed in the Black–Scholes model. In particular, market prices often imply distributions with:

- heavier tails(higher kurtosis),
- negative skewness (higher probability of large downward moves),
- greater probability of extreme events.

These features lead to higher prices for deep out-of-the-money options, especially puts used for downside protection.

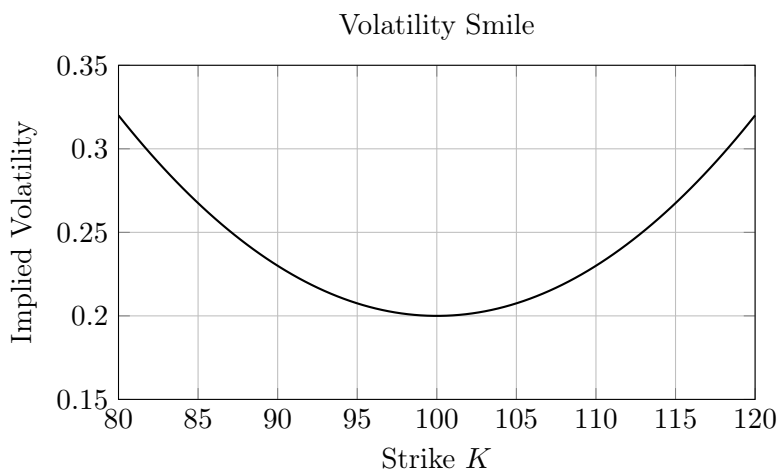

Figure 2: Illustration of a volatility smile where implied volatility varies with the strike price.

11.8 Dupire Local Volatility Formula

The volatility smile observed in option markets indicates that the constant volatility assumption of the Black–Scholes model is inconsistent with market data. One approach to address this is to allow volatility to depend on both the asset price and time:

$$\sigma = \sigma(S, t)$$

This leads to the **local volatility model**

$$dS_t = rS_t dt + \sigma(S_t, t) S_t dW_t.$$

The key question is whether the local volatility function can be determined from observed market option prices. Remarkably, it can be extracted directly from the surface of call option prices across strikes and maturities. This result is known as the **Dupire formula**, derived by *Bruno Dupire*.

Let $C(K, T)$ denote the price of a European call option as a function of strike K and maturity T. Then the local volatility is given by

$$\sigma^2(K, T) = \frac{\frac{\partial C}{\partial T} + rK\frac{\partial C}{\partial K}}{\frac{1}{2}K^2\frac{\partial^2 C}{\partial K^2}}.$$

Using the Breeden–Litzenberger result

$$\frac{\partial^2 C}{\partial K^2} = e^{-rT} f_{S_T}(K),$$

the denominator is proportional to the risk–neutral density of the underlying asset price.

Thus the local volatility surface can be recovered entirely from observed market option prices. In practice, the derivatives of $C(K, T)$ are computed numerically from a smooth interpolation of market option prices.

The Dupire formula provides a method for constructing a volatility surface that exactly reproduces market prices of European options.

Remark 11.1. *In the Black–Scholes framework volatility is assumed to be constant. Given a market option price C_{mkt}, the Black–Scholes formula*

$$C_{BS}(S, K, T, \sigma)$$

can be inverted numerically (for example using the Newton–Raphson or Brent method) to obtain the implied volatility σ_{imp}.

If the Black–Scholes assumptions were correct, the implied volatility would be the same for all strikes and maturities, i.e.

$$\sigma_{imp}(K, T) = \sigma.$$

However, empirical market data shows that implied volatility varies systematically with the strike and maturity. When plotted against the strike price, the resulting curve typically exhibits a smile or skew shape. This phenomenon became particularly evident after the 1987 market crash, when deep out-of-the-money put options were observed to trade at significantly higher prices than predicted by the Black–Scholes model.

This observation indicates that the lognormal distribution of asset prices assumed by Black–Scholes is inconsistent with market prices. In particular, the market-implied distribution tends to exhibit heavier tails and negative skewness.

Using the Breeden–Litzenberger result

$$\frac{\partial^2 C}{\partial K^2} = e^{-rT} f_{S_T}(K),$$

we see that option prices encode the entire risk–neutral probability distribution of the underlying asset. The Dupire formula then provides a method for recovering a local volatility function $\sigma(S, t)$ directly from the observed option price surface $C(K, T)$, allowing the model to exactly reproduce market prices of European options. Also the local volatility model also fails as we still think that volatility is deterministic but in reality its a random variable and thats the foundation of advanced models like Heston model.

12 Derivation of the Brenner-Subrahmanyam Implied Volatility Formula

This section outlines the derivation of the implied volatility formula for at-the-money forward (ATMF) European options, as independently developed by Brenner and Subrahmanyam (1988) and Feinstein (1988).

12.1 Black-Scholes Model and ATMF Condition

The Black-Scholes option pricing model for a European call option on a non-dividend paying stock is defined as:

$$c = SN(d_1) - Xe^{-rT}N(d_2)$$

where c is the call premium, S is the spot price of the underlying asset, X is the strike price, r is the continuously compounded risk-free rate, T is the time left to maturity, and $N(\cdot)$ is the cumulative normal distribution function. The variables d_1 and d_2 are given by:

$$d_1 = \frac{\ln(S/Xe^{-rT})}{\sigma\sqrt{T}} + \frac{1}{2}\sigma\sqrt{T}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

where σ represents the standard deviation of the rate of return on the underlying asset (the volatility).

An option is strictly at-the-money forward (ATMF) when the present value of the strike price payable at maturity exactly equals the current stock price, meaning $S = Xe^{-rT}$. Applying this ATMF condition into the standard Black-Scholes formula simplifies the expressions significantly:

$$c = S[N(d_1) - N(d_2)]$$

Because the fraction S/Xe^{-rT} becomes 1, the natural logarithm term $\ln(1)$ evaluates to zero. Consequently, d_1 and d_2 reduce to:

$$d_1 = \frac{1}{2}\sigma\sqrt{T}$$

$$d_2 = -\frac{1}{2}\sigma\sqrt{T}$$

12.2 Linear Approximation of the Cumulative Normal Distribution

To isolate the volatility parameter σ, the derivation utilizes a linear approximation of the cumulative normal distribution function, which holds for small values of x:

$$N(x) \approx \frac{1}{2} + \frac{1}{\sqrt{2\pi}}x$$

By applying this linear approximation to evaluate $N(d_1)$ and $N(d_2)$ and substituting the differences back into the simplified call option pricing equation, the call price can be estimated as:

$$c \approx \frac{S\sigma\sqrt{T}}{\sqrt{2\pi}}$$

Rearranging this expression to make the implied volatility (σ) the subject provides the Brenner-Subrahmanyam formula for an ATMF call option:

$$\boxed{\sigma \approx \sqrt{\frac{2\pi}{T}}\left(\frac{C}{S}\right)}$$

This demonstrates that for strictly ATMF options, implied volatility is proportional to the ratio of the option price to the stock price.

12.3 Extension to Puts and Straddles

This derivation can easily be extended to put options. Using the put-call parity relationship for European options ($c + Xe^{-rT} = p + S$), it can be shown that when an option is ATMF ($S = Xe^{-rT}$), the call and put premiums are identical ($c = p$). Therefore, the implied volatility can also be calculated using the put option premium:

$$\sigma \approx \frac{p\sqrt{2\pi}}{S\sqrt{T}}$$

In practice, because prices of underlying assets are discrete, the rigid ATMF condition ($S = Xe^{-rT}$) is rarely met exactly. Using the individual call or put formulas in these cases is restrictive and can lead to significant estimation errors, even for near-the-money options.

To reduce these estimation errors, Brenner and Subrahmanyam suggested using a straddle, denoted as z, which is the sum of a call and a put option having the same strike price ($z = c + p$). For an ATMF straddle, the value is simply $2c$, which leads to:

$$z = c + p \approx \frac{2S\sigma\sqrt{T}}{\sqrt{2\pi}}$$

Rearranging this equation to solve for volatility yields the final Brenner-Subrahmanyam straddle formula:

$$\sigma \approx \frac{z\sqrt{2\pi}}{2S\sqrt{T}}$$

For options that are near-the-money (where a call is slightly in-the-money and the corresponding put is out-of-the-money, or vice versa), the arithmetic average provided by the straddle helps offset errors, making this formula highly accurate for calculating implied volatility.

- **Taylor Expansion of $N(x)$:** The linear approximation $N(x) \approx \frac{1}{2} + \frac{1}{\sqrt{2\pi}}x$ is derived using a first-order Taylor series expansion of the standard normal cumulative distribution function (CDF) around $x = 0$. The Taylor expansion formula is $N(x) \approx N(0) + N'(0)(x - 0)$. We know that $N(0) = 0.5$, as exactly half the area under the standard normal curve lies below zero. The first derivative of the CDF, $N'(x)$, is the standard normal probability density function (PDF), $n(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$. Evaluating this derivative at $x = 0$ gives $N'(0) = \frac{1}{\sqrt{2\pi}}$. .
- **ATMF Options, Forward Price, and Dividends:** An At-The-Money Forward (ATMF) option is one where the strike price (X) is set equal to the theoretical forward price of the underlying asset at the option's expiration date. The **forward price** is the predetermined delivery price of an asset agreed upon today for a transaction that will occur at a future date. For a non-dividend paying stock, the forward price is calculated by growing the current spot price (S) at the risk-free rate (r) over the time to maturity (T), yielding $F = Se^{rT}$. A **dividend** is a distribution of a portion of a company's earnings to its shareholders. In option pricing, if a stock pays a continuous dividend yield (q), the forward price adjusts downward to $F = Se^{(r-q)T}$ because the option holder does not receive these cash distributions. Because the strike price is a cash flow that will be exchanged in the future (at time T), it must be discounted back to the present to be meaningfully compared with today's stock price. The discounted present value of the strike is Xe^{-rT}. Therefore, an option is ATMF when the current stock price perfectly matches the present value of the strike price ($S = Xe^{-rT}$). Economically, this means investing S dollars today at the risk-free rate will yield exactly the strike price X at maturity.